Electric fields

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use field arrows to predict the force on a test charge.

SUCCESS CRITERIA - I can...

1. Draw electric field arrows for isolated positive and negative charges.
2. Use the positive-test-charge convention.
3. Distinguish the field from a particular test charge's force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "electric field" and "charge" to explain your model or result. Sentence starters are optional.

electric field: force per positive test charge at a location

charge: a property responsible for electric interactions

vector: a quantity with size and direction

01 Warm-up: bring back an earlier idea

In separate cases, an ideal blackbody temperature becomes 3 times as large and a sound frequency becomes 5 times as large at fixed sound speed. **Compare** the two wavelength changes.

Word help: wavelength = distance between corresponding points on repeated waves

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: E = F/q; F = qE.
Use consistent signed components. The field points along the force on a positive test charge; negative charge reverses force direction.

Worked reasoning: At a point where E=4 N/C right, a +2 C test charge feels 8 N right. A-2 C charge feels 8 N left. Field direction is defined using positive test charge.

Your turn: Draw the field arrow once and the two alternative force arrows.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Field / first contribution

Force / second contribution

Draw separate arrows; label quantities and units.

Marzano check - circle: ① ② ③

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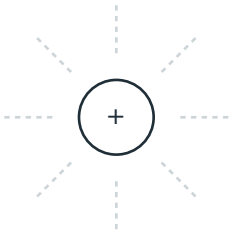
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Sketch field arrows around a positive source charge. Place a negative test charge to its right and draw that charge's force.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Add arrowheads to show field direction.

The center marks the source; add arrowheads on the surrounding lines to show field direction.

Marzano check - circle: 1 2 3

04 Apply the physics

A tiny charged bead has charge -2.0 × 10^{-6} C in a uniform electric field of 300 N/C east.
Calculate its electric force.
Use separate arrows to explain why force and field point in different directions.

Word help: uniform = constant throughout the stated interval

Sentence starter: Both ... . They differ in ... because ... .

Field / first contribution

Force / second contribution

Draw separate arrows; label quantities and units.

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

If you remove the test charge, does the source charge's field disappear?
Explain the role of a test charge.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A closed-open pipe has length 0.4 m.
Use λ=4 L for the fundamental and v=340 m/s.
Find its fundamental frequency.
Sketch the displacement pattern and label the pipe length.

Sentence starter: I found … [unit]. This value means … in this situation.

closed-open tube

Draw a displacement pattern; label nodes and antinodes.

Marzano check - circle: ① ② ③

07 Exit check

At twice the distance from a point charge, how does field magnitude change?
State the model assumption.

Word help: magnitude = size of a quantity without its direction

Sentence starter: Under the assumption …, the evidence … supports … because … .

Field / first contribution

Force / second contribution

Draw separate arrows; label quantities and units.

Marzano check - circle: ① ② ③

08 STRETCH: test the model

A -3 microcoulomb charge feels a 0.012 N force upward.
Find the electric-field vector.

Sentence starter: I found … [unit]. This value means … in this situation.

Field / first contribution

Force / second contribution

Draw separate arrows; label quantities and units.

Marzano check - circle: ① ② ③